

EXPERIENCE IN DEVELOPING DIGITAL ECONOMY IN SOME COUNTRIES AND LESSONS FOR VIETNAM

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Abstract: *Developing the digital economy involves leveraging digital technology and data to create innovative business models that transform traditional production processes into ecosystem-based operations. This integration enhances productivity and responds effectively to consumer needs. Learning from the experiences of developed countries is crucial for Vietnam as it formulates policies for its own digital economy amidst global integration. This article analyzes international digital economy development experiences, assesses Vietnam's current status, identifies existing advantages and challenges, and recommends policies to foster the growth of Vietnam's digital economy.*

• Keywords: : digital economy, international experience, e-commerce, economic development.

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1. Introduction

In today's era, the digital economy is not just a trend but a force that is completely changing the way we work, communicate and consume. From the explosion of the internet to the spread of artificial intelligence and blockchain, the digital economy has created a wave of innovation and constant transformation. In this context, it is extremely important to grasp and understand the trends, challenges and opportunities of the digital economy.

The digital economy is a concept that refers to the use of digital technology to create and distribute economic value. It includes the widespread application of information and communication technology (ICT) in all aspects of business operations, from business management to production, marketing, delivery and customer service. Key elements of the digital economy include:

Information and communications technology (ICT): Includes the internet, computers, mobile phones, business applications and software to create and share information, manage production and distribution processes, and interact with customers.

E-commerce: An important part of the digital economy, e-commerce allows the purchase and sale of goods and services over the internet or other online platforms without the need for face-to-face meetings.

FinTech: Includes convenient financial services such as online payments, online loans, personal finance management and other technology-based financial services.

Artificial Intelligence (AI): AI plays an important role in generating analytical and predictive information, improving business process automation and enhancing customer experience through chatbots and automated support systems.

Blockchain: Blockchain technology is used to authenticate and record transactions in a transparent and secure manner, especially in the financial and supply chain sectors.

The digital economy is not simply the use of digital technology, but also proposes a new approach to creating value and improving performance in the global economy.

In recent decades, Vietnam has witnessed rapid growth in the digital economy, especially in the following areas:

According to a report by the Ministry of Industry and Trade, in 2023, the size of the retail e-commerce market will increase by about 4 billion USD (up 25%) compared to 2022, reaching 20.5 billion USD. It is forecasted that in the period 2022 - 2025, Vietnam's Internet economy will have a leading growth rate in the region at about 31%/year. According to Modor Intelligence, in January 2024, Vietnam was recognized as one of the 10 countries with the highest e-commerce growth rate in the world, leading Southeast Asia. According to Modor Intelligence, the size of Vietnam's e-commerce market is estimated to reach 14.7 billion USD in 2024 and is expected to reach 23.77 billion USD in 2029, growing at a CAGR of 10.09% during the forecast period (2024-2029).

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The value of investment deals in the Fintech sector in Vietnam reached 294 million USD. According to the survey and synthesis information of HyperLead, in terms of the number of transactions, Vietnamese Fintech startups have received about 14 investments, the leading affiliate marketing platform in Vietnam specializing in providing products and services in the field of Finance - Banking - Fintech, the number of Startups has increased by nearly 13% (from 156 companies in 2021 to 176 companies in 2022). In which, Payment is still the most vibrant Fintech Startup field in Vietnam, accounting for 22.6% of the number of Fintech companies in Vietnam, followed by Personal Lending and Blockchain/Crypto.

Vietnam is becoming one of the countries with strong potential for digital economic development, with support from the development of information technology infrastructure and the government's strategic vision of promoting the Industrial Revolution 4.0.

In addition to the introduction presented above, the rest of the study is presented as follows: In the next part, the study conducts the review on advantages and difficulties in developing digital economy in Vietnam, then the study proceeds experience in developing digital economy in some countries around the world. Finally, the study proposes reference lessons for developing digital economy in Vietnam.

2. Advantages and difficulties in developing digital economy in Vietnam

2.1. Advantages in developing digital economy in Vietnam

Young population and number of internet users: With a significant youth demographic and a growing number of internet users, Vietnam has emerged as a substantial market for digital economic offerings. Statistics indicate that approximately 72% of the population owns smartphones, and 68% engage with videos and music daily via mobile devices. Furthermore, around 70% of mobile subscribers utilize 3G or 4G networks, indicating robust mobile connectivity.

Rapid growth of the e-commerce market: The proliferation of smartphones and internet access has spurred a flourishing e-commerce landscape. Vietnam is home to about 30,000 technology companies, including 10,000 rapidly expanding software firms. The market features over 50 financial technology companies and transportation apps like Grab and Uber, alongside tourism startups such

as Mytour and Luxstay that compete with global platforms. Annually, thousands of innovative digital startups emerge, many successfully executing high-tech initiatives.

Government support policies: The Vietnamese government has rolled out various supportive policies for the digital economy, focusing on the 4.0 industrial revolution and fostering technology startups and national digital platforms. The Prime Minister has endorsed the National Digital Transformation Program for 2025 and the e-commerce development Master Plan for 2021-2025, emphasizing that digital transformation is crucial for Vietnam's development. These initiatives aim to modernize distribution systems, enhance enterprise competitiveness, and stimulate both domestic market growth and exports.

Development of the information and communication technology (ICT) industry: Vietnam's ICT sector is rapidly advancing, featuring established technology enterprises and research and development centers. By the end of 2020, major mobile operators such as Viettel, VNPT, and Mobifone introduced 5G services, positioning Vietnam among the early adopters of this technology globally. The country is transitioning from a reliance on imported technologies to mastering and producing 5G equipment, marking a strategic leap in its telecommunications landscape.

Labor potential: Vietnam's labor force is characterized by low costs and a capacity for rapid adaptation to new technologies, providing favorable conditions for businesses to implement digital economic projects effectively.

Growing consumer market: The rise in income levels and the expanding middle class in Vietnam are driving demand for digital products and services, further fueling the growth of the digital economy.

2.2. Difficulties in developing digital economy in Vietnam

Information technology infrastructure: Despite rapid development, information and communications technology (ICT) infrastructure in Vietnam is still limited in some areas, especially in rural and mountainous areas.

Cybersecurity and information security: Cybersecurity and information security issues are a major challenge to the development of the digital economy in Vietnam, such as the risk of cyberattacks, information security violations and online crimes. Cybersecurity measures need to be strengthened to ensure the safety of data and online transactions.

Access and digital divide: Despite the increase in internet and smartphone access, there are still disparities in access to digital technology between regions and social classes in Vietnam. The infrastructure for the digital economy is not synchronous, a common national database has not been built; the logistics system is weak; the rate of enterprises applying digital technology and software is still very low, of which 59% is for human resource management applications, 29% for supply chain management, and 32% for customer relations.

Human resource development: To promote the digital economy, Vietnam has invested in developing highly skilled human resources in the field of information and communication technology, as well as in applying digital technology to other business areas. However, the shortage of quality human resources in this field still exists. Training and attracting talents with high professional knowledge and creativity remains a major challenge.

Policy and legal regulations: The institutional and legal environment for digital economic development is still loose, inconsistent, lacking transparency and creativity.

3. Experience in developing digital economy in some countries around the world

Most countries worldwide, both developed and developing, prioritize the development of the digital economy as a crucial step for future economic models and transformation. Each nation, however, employs distinct mechanisms, policies, and measures tailored to its unique characteristics and strengths. The following section highlights the experiences of various countries in developing their digital economies, focusing on examples from Southeast Asia and beyond.

Germany: Germany is advancing its digital economy through a comprehensive strategy encompassing national, industry, and enterprise levels. The government has established the Industry 4.0 Department within the Ministry of Economic Affairs and Energy to oversee digital transformation, supported by specialized working groups focused on areas like standards development, training, and cybersecurity. Leveraging its strengths in heavy industrial production, Germany emphasizes the application of digital technologies in emerging industries such as robotics, smart vehicles, and smart energy, while also transforming human resource training to promote flexibility and digital

skills through collaboration between educational institutions and businesses. Additionally, Germany is developing smart manufacturing systems and factories that utilize IoT and automation, linking all production processes to operate cohesively as a large smart factory.

Estonia: Estonia has effectively advanced its digital economy through initiatives like the “Estonian Broadband Infrastructure - Network Project” (EstWin), launched in 2009 to install 6,600 km of fiber optic cables in rural areas. By 2020, 98% of households and organizations were within 1.5 km of EstWin, with 85% of the funding sourced from the European Regional Development Fund and the remainder from network operators. Additionally, the Estonian government has implemented programs such as Digital Agenda 2020 and the Estonian Lifelong Learning Strategy 2020, creating a robust digital information infrastructure (X-road) for citizens to access services like electronic voting and online healthcare. These efforts have been complemented by a focus on enhancing human resources, highlighted by the “Tiger Leap” initiative, which has helped Estonia rank 10th in human resources performance among EU countries, emphasizing the importance of digital skills for sustainable economic growth.

Japan: Japan’s digital economic development program is integrated into its broader Society 5.0 initiative, featuring several key strategies and policies. These include an investment strategy aimed at restructuring economic sectors and employment to adapt to the digital economy, enhancing information security infrastructure, and promoting IT usage across various sectors. Additionally, Japan seeks to become the world’s leading information technology nation, positioning IT as a cornerstone of its growth strategy, while also implementing a Cybersecurity Strategy that focuses on leveraging technological advances and developing skilled human resources to effectively combat cyber threats.

Korea: Korea recognizes the digital economy and the 4.0 industrial revolution as key drivers of future economic development, prioritizing these sectors in its economic policies. The government is actively promoting science, technology, and innovation, enhancing the authority of the Science and Technology Innovation Agency (STI) and increasing budget allocations for research and development. Additionally, a Fourth Industrial Revolution Committee has been established to strategize for future

industries, while comprehensive interdisciplinary development is emphasized across various sectors, particularly in industry and information technology. Supportive policies have been implemented to foster new technology projects, reduce risks for innovative startups, and invest in high-quality human resources, facilitating the advancement of the digital economy in Korea.

China: The development of China's digital economy is significantly driven by supportive policies and strategic planning aimed at fostering domestic digital technology. China prioritizes the protection of its domestic digital market, allowing local enterprises to establish and maintain a strong presence while supporting advancements in science, technology, and innovation within the digital sector. Initial efforts focus on moderately complex sectors like e-commerce, gradually progressing to more advanced fields such as artificial intelligence and robotics. By banning Western tech giants like Meta and Google, China simultaneously nurtures domestic firms such as Weibo, Baidu, and Bytedance, enabling them to develop new technologies and patents. This approach has cultivated a comprehensive digital ecosystem encompassing production, transportation, trade, and payment, optimizing national resources and facilitating economic growth. Moreover, the government emphasizes the transition to digital methods and transactions, promoting initiatives in e-government, e-banking, and e-commerce.

Singapore: Singapore aspires to become a global leader in the digital economy by implementing comprehensive strategies that support digital transformation across various industries. This includes financial assistance for businesses to develop digital capabilities, with funding incorporated into the government's annual budget, along with ongoing advisory support to tailor digitalization plans. To foster human resource development, Singapore emphasizes training programs for digital skills, particularly targeting unskilled workers and those at risk of job displacement due to digital changes. Additionally, the education system integrates digital skills into curricula, covering areas such as coding and design thinking. To create a secure environment for its digital economy, Singapore has enacted the Cybersecurity Law 2018 and the Personal Data Protection Law, enhancing protections for critical information infrastructure and personal data while establishing a legal framework for data management.

Malaysia: Malaysia has taken significant steps to enhance its digital economy, being the first Southeast Asian country to enact the Data Protection Act and implementing various infrastructure projects, including the High-Speed Broadband (HSBB) initiative aimed at expanding broadband access. The government has launched programs like the "Malaysia Tech Entrepreneur Program" (MTEP) to attract foreign direct investment by offering tax incentives and visa opportunities for tech entrepreneurs. Additionally, Malaysia emphasizes e-commerce development in alignment with new-generation trade agreements and focuses on digital transformation in public services through the Malaysian Administrative Management and Modernization Unit (MAMPU), which aims to improve service delivery using information and communication technology. The country also invests in workforce training programs, such as eUshawan and eRezeki, to equip young entrepreneurs and create job opportunities for digital workers across various sectors.

Thailand: Thailand has created a comprehensive legal framework to support digital economic development by consolidating three existing laws and drafting eight new regulations, including the Cybersecurity Law and the Personal Data Protection Law. The Digital Development Law establishes the Digital Economy Commission and the Digital Economy and Society Promotion Agency, tasked with fostering digital industry growth and technology application for economic benefits. Additionally, Thailand has set up the Digital Economy and Society Development Fund to enhance telecommunications infrastructure, funded by the government's budget and licensing fees, while prioritizing network security and content regulation in cyberspace.

4. Reference lessons for developing digital economy in Vietnam

From the analysis of the experiences of some European countries (Germany, Estonia), Asian countries (Japan, Korea, China, Malaysia, Singapore, Thailand) on digital economic development (strategy, policy) above, it is possible to draw some policy implications to apply to develop the digital economy in Vietnam in the period 2024 - 2030:

First, perfecting institutions and policies on digital economic development

Vietnam should prioritize developing and refining institutions and policies to establish a robust

framework for digital economic growth across provinces and cities. Enhancing these frameworks will boost local capabilities to attract investments in digital technology and venture capital, streamline processes for capital contributions, and facilitate mergers and acquisitions in the tech sector. Additionally, Vietnam must create specific digital transformation plans for businesses, ensuring these are linked to financial resources and annual budgets, while also addressing potential negative impacts of digitalization, such as job displacement and consumer rights protection, to foster a balanced and secure digital economy.

Second, clearly identify breakthrough industries and fields to develop the digital economy.

Vietnam needs to base on the strengths and characteristics of each locality to clearly identify breakthrough industries and fields to develop the digital economy. In the immediate future, provinces and cities should focus on developing digital economic fields to help expand the market and promote consumption because these fields help localities take advantage of opportunities from international integration, especially the new generation free trade agreements that Vietnam has signed (CPTPP, EVFTA, EVIPA) but also do not require high technology level. At the same time, provinces and cities should develop the fundamental fields of the digital economy, such as digital infrastructure, digital resources, digital services, digital markets and have a methodical plan and strategy to access and master important, core technologies of the digital economy through support policies and in cooperation and attracting foreign investment.

Third, promote awareness of the digital economy in society.

Vietnam needs to focus on propaganda activities to raise awareness among people and businesses about the digital economy, its benefits and challenges, with specific content for each industry and field, especially through agencies, businesses and schools. At the same time, press and media agencies need to guide public opinion, helping businesses, people and the whole society have a correct understanding of the digital economy, thereby being best prepared to adapt to the above development trend. This measure also helps to enhance the responsibility of the Government, businesses and people in building the digital economy.

Fourth, focus on developing human resources for the digital economy.

This issue has long been focused on by developed countries, but for developing countries like Vietnam, the issue of human resource development for the digital economy has not received due attention. Therefore, Vietnam needs to devote more resources to human resource development for the digital economy. In particular, it should focus on developing and attracting digital technology experts and digital entrepreneurs through appropriate policies. At the same time, it is necessary to orient the development of local education and training in conjunction with the socio-economic development strategy and the development orientation of high-tech industries and fields, smart manufacturing of the knowledge economy; innovate and modernize university and vocational training programs in provinces and cities, strongly shifting from training based on ability to training based on social needs.

Fifth, focus on investing in infrastructure in the process of developing the digital economy.

Infrastructure is crucial for advancing Vietnam's digital economy; thus, significant investment in upgrading technology and mastering core technologies is essential. The central government should collaborate with localities to develop a cohesive national digital infrastructure that supports data connection, storage, and processing while ensuring network security. This includes enhancing broadband infrastructure, upgrading 4G networks, and rolling out 5G to provide universal access to high-speed internet, cloud computing services, and advanced technologies for all citizens, while also encouraging participation from various economic sectors in infrastructure investment, with careful attention to national security concerns.

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